

REVIEW ARTICLES
SYSTEMATIC REVIEWS

Combining ACL Reconstruction with Lateral Extra-Articular Tenodesis Reduces Long-Term Osteoarthritis Risk Versus Isolated ACL Reconstruction

A Systematic Review and Meta-Analysis

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Background: Lateral extra-articular tenodesis (LET) is increasingly used to augment anterior cruciate ligament reconstruction (ACLR), particularly in patients with high-grade rotatory instability. Despite demonstrated biomechanical advantages, the long-term effect of LET on posttraumatic osteoarthritis (OA) remains unclear.

Methods: We hypothesized that adding LET to ACLR reduces development of long-term moderate-to-severe radiographic OA, particularly after meniscectomy. We performed a PRISMA-compliant systematic review and meta-analysis, which was registered with PROSPERO. PubMed, Cochrane CENTRAL, and Scopus were searched through March 2025. Eligible randomized controlled trials (RCTs) or comparative cohort studies reported radiographic moderate-to-severe OA after ACLR with versus without LET, with a minimum follow-up of 5 years. Pooled odds ratios (ORs) with 95% confidence intervals (CIs) were calculated using a random-effects model.

Results: Six studies with 444 patients (mean follow-up, 13.1 years) were included. LET was associated with a significant reduction in moderate-to-severe OA in the lateral compartment. Subgroup analysis showed a significant reduction in lateral OA as measured using both the Kellgren-Lawrence (K-L) (OR, 2.87; 95% CI, 1.2 to 6.9; $p = 0.02$) and International Knee Documentation Committee (IKDC) classifications (OR, 4.38; 95% CI, 1.5 to 12.7; $p = 0.01$). In contrast, no significant difference was found in the medial compartment for either the K-L (OR, 1.26; 95% CI, 0.7 to 2.4; $p = 0.49$) or IKDC classification (OR, 2.06; 95% CI, 0.6 to 6.8; $p = 0.24$). In meniscectomized knees, LET significantly reduced OA risk, especially as measured with the IKDC (OR, 6.14; 95% CI, 1.7 to 22.6; $p = 0.01$) compared with the K-L classification (OR, 3.61; 95% CI, 1.1 to 12.3; $p = 0.04$). In contrast, although LET also reduced OA risk in non-meniscectomized knees, the difference was not significant. Compartment-specific OA data were unavailable within the meniscal subgroups.

Conclusions: This meta-analysis, the first to assess the long-term effect of LET on OA, indicated that LET significantly reduces moderate-to-severe OA risk, particularly in the lateral compartment and among meniscectomized knees. Findings support selective LET use during ACL reconstruction, especially in cases with compromised meniscal integrity, to help mitigate OA progression.

Level of Evidence: Therapeutic Level II. See Instructions for Authors for a complete description of levels of evidence.

Management of anterior cruciate ligament (ACL) injuries has advanced with improved understanding of knee anatomy, biomechanics, and surgical techniques¹⁻³. Residual rotational laxity after ACL reconstruction

(ACLR) is linked to anterolateral complex insufficiency, underscoring the need to address these structures⁴. Meniscal injuries, especially posterolateral root tears, exacerbate instability and severe pivot shift in ACL-deficient knees^{5,6}.

Disclosure: The Disclosure of Potential Conflicts of Interest forms are provided with the online version of the article (<http://links.lww.com/JBJS/I987>).

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Lateral extra-articular tenodesis (LET), formerly a standalone procedure, is now an adjunct to ACLR, particularly in patients with high-grade rotatory instability. Recent reviews show that LET reduces pivot shift, improves rotational stability and clinical outcomes, and lowers the rate of ACL graft failure^{3,7,8}.

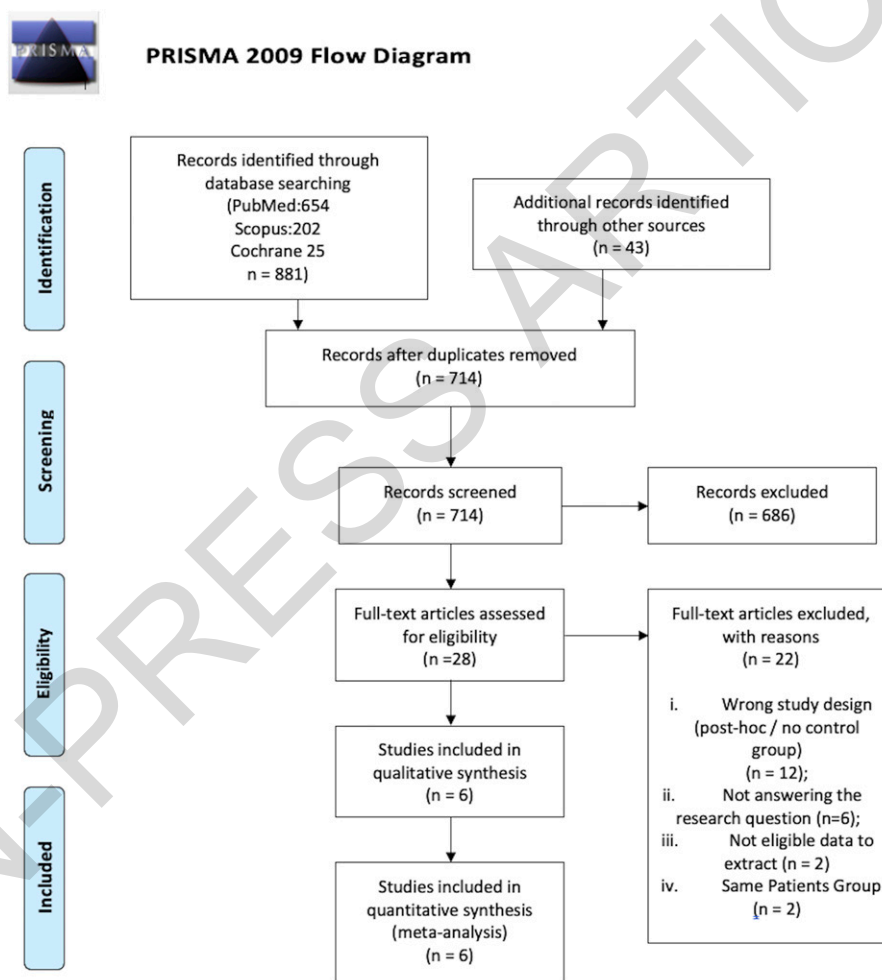
ACL injury increases osteoarthritis (OA) risk over time, regardless of treatment^{9,10}. Although LET improves stability and reduces graft failure, its long-term effect on OA progression remains unclear. Concerns persist that LET may over-tighten the lateral compartment, alter biomechanics, and increase OA risk^{1,11}. Consequently, clarifying whether LET delays or exacerbates OA is essential.

To our knowledge, no meta-analysis has assessed long-term radiographic OA after ACLR with versus without LET. We

aimed to evaluate long-term radiographic OA in patients undergoing ACLR with versus without LET at a minimum follow-up of 5 years. We hypothesized that (1) LET reduces the risk of moderate-to-severe OA, and (2) this effect is greater in patients with concomitant meniscectomy.

Materials and Methods

This PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)-compliant study¹² was registered in PROSPERO (CRD420251050747). Figure 1 shows the flow diagram; the PRISMA checklist appears in Appendix Supplementary Table 1. The eligibility criteria, data extraction, and statistical analysis plan were predefined.



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For more information, visit www.prisma-statement.org.

Fig. 1

PRISMA flowchart for the search and identification of included studies.

Search Strategy

We searched PubMed, Cochrane CENTRAL, and Scopus from inception through March 31, 2025. Grey literature was screened via OpenGrey, the WHO (World Health Organization) International Clinical Trials Registry Platform, the ISRCTN, and ClinicalTrials.gov. The PICO (Population, Intervention, Comparison, Outcome)-based strategy targeted studies of ACLR with versus without LET that reported radiographic OA outcomes. Two independent teams screened records; disagreements were resolved by consensus with a third reviewer. A representative search string appears in Appendix Supplementary Table 2.

Trial Selection

Eligible peer-reviewed trials or comparative cohort studies reported long-term (minimum 5-year) outcomes after ACLR with versus without LET, and provided radiographic data specific to moderate-to-severe OA. Studies in English, German, or French with extractable data were included. Studies were excluded if they were post-hoc analyses; lacked a control group; provided non-extractable data; were review articles, technical notes, biomechanical or animal studies; or were in other languages.

Data Synthesis and Analysis

Four reviewers (N.K.G., D.S., A.A.K., and I.C.) independently screened and extracted data. Variables included first author and year, country, design, follow-up, demographics, group sizes, ACL graft type, LET technique, and radiographic classification. OA severity in the studies was classified using the Kellgren-Lawrence (K-L), International Knee Documentation Committee (IKDC), Fairbank, and/or Ahlbäck systems¹³⁻¹⁶. For consistency in analysis, OA grades were standardized to normal-to-mild versus moderate-to-severe¹. Meniscectomy status was recorded. Data were summarized as the means or medians and ranges or as percentages.

Risk of Bias and Study Quality Assessment

The risk of bias in randomized controlled trials (RCTs) was assessed using the Cochrane RoB 2.0 tool across 5 domains: bias arising from the randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. The risk of bias in cohort studies was assessed using the Newcastle-Ottawa Scale, whose domains are selection, comparability, and exposure/outcome.

Statistical Analysis

Analyses used random-effects models to account for heterogeneity if the analysis included >2 studies, or a fixed-effects

model if the analysis included only 2 studies. Odds ratios (ORs) with 95% confidence intervals (CIs) were calculated for the primary outcome of the development of moderate-to-severe

OA as seen on radiographs. Heterogeneity was assessed using I^2 and Cochrane χ^2 values; an I^2 of >60% indicated substantial heterogeneity. Subgroup analyses were performed on the basis of the compartment (OA in the lateral compartment and in the medial compartment), the OA grading system used (K-L and IKDC), and meniscal status. Sensitivity analyses evaluated the effects of the study design, geographic origin of the study, graft type, and LET technique. As there were only 6 studies, publication bias was not tested statistically, but a funnel plot is provided in Appendix Supplementary Figure

1. All tests were 2-sided; $p < 0.05$ was considered significant. Analyses were performed using IBM SPSS Statistics, version 29.

Results

Descriptive Characteristics of the Included Studies

We screened 924 articles, of which 28 underwent full-text review. Twenty-two were excluded due to an ineligible design ($n = 12$), irrelevant outcome ($n = 6$), insufficient data ($n = 2$), or patient overlap ($n = 2$). Across the remaining 6 included studies, 444 patients contributed to the target comparison (ACLR+LET versus isolated ACLR)¹⁷⁻²² (Fig. 1). Mean follow-up was 13.1 years (range, 5 to 20 years). Baseline characteristics were comparable except in the study by Castoldi et al.¹⁸, which was assessed as high-risk according to the RoB 2.0 tool and excluded in a sensitivity analysis due to meniscal status imbalance (Table 1; see also Appendix Supplementary Table 3).

OA Prevalence with Versus without LET in Pooled Classification Systems

Pooled analysis of all 6 studies initially showed no significant difference in OA prevalence between the ACLR and ACLR+LET groups in either the lateral compartment (OR, 1.86; 95% CI, 0.5 to 6.9; $p = 0.35$) or medial compartment (OR, 1.48; 95% CI, 0.8 to 2.7; $p = 0.21$), with moderate heterogeneity ($I^2 = 47\%$) (Fig. 2-A). The funnel plot for qualitatively assessing publication bias is shown in Appendix Supplementary Figure 1. Visual inspection of the funnel plot shows approximate symmetry, suggesting no strong evidence of publication bias; given the small number of studies, formal tests would have been underpowered and were not performed. However, a sensitivity analysis was performed excluding the study by Castoldi et al.¹⁸. In that study, which reported a mean follow-up of 19.4 years, there was a significant imbalance in the proportion of meniscectomized knees: 11 of 22 in the ACLR+LET group

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Findings support selective LET use during ACL reconstruction, especially in cases with compromised meniscal integrity, to help mitigate OA progression.”

Table 1. Descriptive Characteristics of the Studies*

Study	Country	Follow-up (yr)	Study Design	ACLR+LET/ACLR				No. of Patients		LET Procedures	OA Classification Systems
				Age at Surgery† (yr)	M:F Sex (%)	Grafts (no.)	Meniscectomized, Yes:No (no.)	Total	In Our Analysis		
Lucidi (2025) ¹⁹	Italy	20	RCT	27.9 ± 7.1/31.3 ± 7.6	68:32/65:35	25 HT/25 HT	No meniscal injury (per inclusion criteria)	75	24	HT over-the-top LET	K-L
Shatrov (2023) ²¹	France	8.6	Cohort	28.4 ± 10.8/28.5 ± 10	71.4:28.6/60.5:39.5	42 HT/29 HT+9 BTB	22:20/19:19	80	80	ALLR	K-L, IKDC
Viglietta (2022) ²²	Italy	15.7	Cohort	26.1 ± 6/28.4 ± 6.4	82.2:17.8/71.7:28.3	76 HT/79 HT	31:45/28:51	165	155	Arnold-Coker‡	K-L, IKDC, Fairbank
Castoldi (2020) ¹⁸	France	19.4	RCT	26.2 ± 5.4/ 25.9 ± 5.18	78.3:21.7/70.5:29.5	22 BTB/23 BTB	11:11/4:19§	120	45	Modified Lemaire	Ahlbäck
Rota (2020) ²⁰	Italy	10	Cohort	28.7 ± 4.2/28.5 ± 3.7	100:0/100:0	20 HT/20 HT	5:15/5:15	60	40	Arnold-Coker‡	K-L, IKDC, Fairbank
Acquitter (2003) ¹⁷	France	5	RCT	27.2/27	68:32/76:24	50 BTB/50 BTB	#	100	100	Extra-articular plasty using a quadriceps tendon strip	IKDC

*ACLR = anterior cruciate ligament reconstruction, LET = lateral extra-articular tenodesis, OA = osteoarthritis, RCT = randomized controlled trial, HT = hamstrings graft, K-L = Kellgren-Lawrence, BTB = bone-patellar tendon-bone graft, ALLR = anterolateral ligament reconstruction, IKDC = International Knee Documentation Committee. †Values are given as the mean with or without the standard deviation. ‡The Arnold-Coker modification of the McIntosh procedure. §The authors reported a significant difference in the percentage of meniscectomized knees between the ACLR and ACLR+LET groups. #Not stated, but the authors reported nonsignificant differences in the percentage of meniscectomized knees between the 2 groups.

compared with 4 of 23 in the isolated ACLR group. This was the only included study with such a discrepancy. After exclusion, a significant effect favoring LET in the lateral compartment was observed (OR, 3.52 for isolated ACLR; 95% CI, 1.4 to 8.8; $p = 0.01$), along with a substantial reduction in heterogeneity ($I^2 = 7\%$), at a revised mean follow-up of 11.9 years (Fig. 2-B). As before, no significant difference was found in the medial compartment (OR, 1.92; 95% CI, 0.7 to 4.9; $p = 0.17$).

Subgroup Analysis by OA Classification System

Analysis of the various OA classification systems individually demonstrated consistent protective effects of LET. For moderate-to-severe OA defined according to the K-L system ($n = 3$), the pooled OR for the lateral compartment was 2.87 (95% CI, 1.2 to 6.9; $p = 0.02$) for isolated ACLR, and the mean follow-up was 13.6 years, based on 3 studies^{19,20,22}. No significant effect was observed in the medial compartment (OR, 1.26; 95% CI, 0.7 to 2.4; $p = 0.49$). The analysis based on the IKDC classification ($n = 4$) demonstrated an even stronger protective effect in the lateral compartment, with an OR of 4.38 for isolated ACLR (95% CI, 1.5 to 12.7; $p = 0.01$). The mean follow-up for this subgroup was 13.6 years, based on 4 studies^{17,20-22}. Again, no significant difference was found in the medial compartment (OR, 2.06; 95% CI, 0.6 to 6.8; $p = 0.24$) (Figs. 3-A and 3-B). Comparative characteristics of the included studies are provided in Appendix Supplementary Table 4.

Subgroup Analysis by Meniscal Status

Two studies provided stratified data on meniscal status that permitted investigating whether the effect of LET on the development of moderate-to-severe OA following ACLR differed according to whether a meniscectomy had been per-

formed^{20,22}. Thus, we conducted subgroup meta-analyses comparing ACLR versus ACLR+LET within the meniscectomized subgroup and, separately, within the non-meniscectomized subgroup. However, compartment-specific OA data were unavailable within the meniscal subgroups. Among meniscectomized knees, those treated with ACLR alone demonstrated a significantly higher risk of developing moderate-to-severe OA compared with those treated with ACLR+LET. This was observed with both the K-L classification (OR, 3.61; 95% CI, 1.1 to 12.3; $p = 0.04$) and the IKDC system (OR, 6.14; 95% CI, 1.7 to 22.6; $p = 0.01$). The mean follow-up across these studies was 12.9 years (Figs. 4-A and 4-B). In contrast, no significant differences were observed in non-meniscectomized knees with either the K-L classification (OR, 2.07; 95% CI, 0.4 to 9.7; $p = 0.35$) or the IKDC classification (OR, 3.57; 95% CI, 0.6 to 22.5; $p = 0.18$). These findings suggest that the protective effect of LET may be more pronounced in patients with meniscal injuries that cannot be repaired.

Additional Sensitivity Analyses

Exploratory sensitivity analyses tested the robustness of the results with respect to risk of bias, study design, geographic origin, ACLR graft option, and LET technique. Notably, when we excluded studies utilizing bone-patellar tendon-bone (BTB) grafts, specifically those by Castoldi et al.¹⁸ and Acquitter et al.¹⁷, the observed protective effect of LET against lateral compartment OA became more pronounced (OR, 5.57; 95% CI, 1.9 to 15.9; $p < 0.001$), while medial compartment results remained nonsignificant.

Discussion

To our knowledge, this study represents the first systematic review and meta-analysis to evaluate the long-term

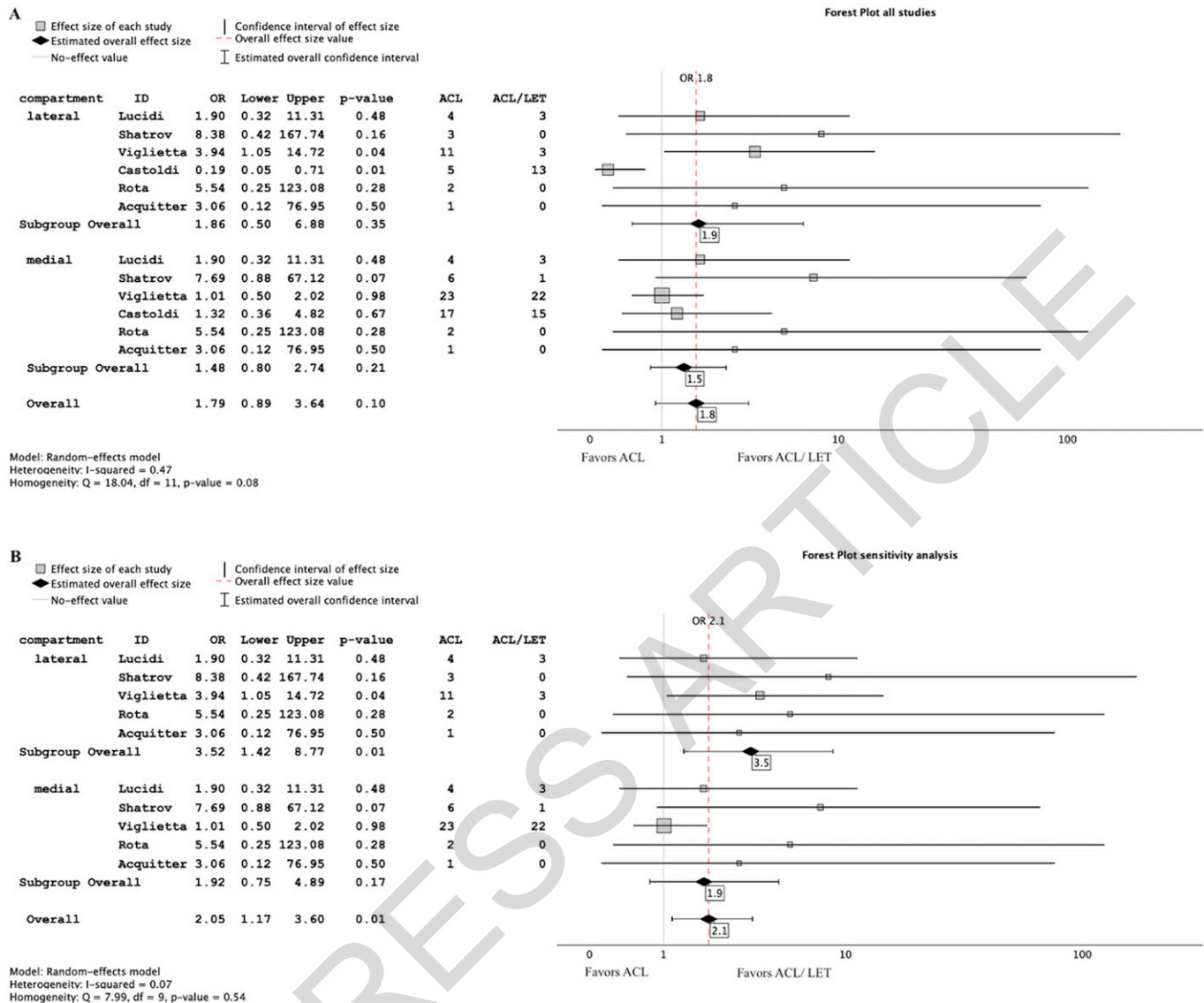


Fig. 2

Forest plots showing the odds ratios (ORs) for the development of moderate-to-severe osteoarthritis (OA) in the lateral and medial compartments after ACLR with versus without LET, based on pooled OA classification systems, in all 6 included studies (**Fig. 2-A**) and in a sensitivity analysis excluding the study by Castoldi et al.¹⁸, which had an imbalance in the proportion of patients with meniscectomy between the groups and a high risk of bias (**Fig. 2-B**). df = degrees of freedom.

radiographic outcomes, focusing on OA development, of ACLR with versus without LET. This meta-analysis suggests that LET is associated with a protective effect against lateral compartment OA progression, particularly in meniscectomized knees.

OA remains an important long-term complication of ACL injuries. Liukkonen et al. reported a posttraumatic OA prevalence of 37.9% at 14.6 years in operatively treated patients versus 40.5% at 15 years in nonoperatively treated patients¹⁰. Factors contributing to OA progression include meniscal injury, cartilage damage, and altered biomechanics. Systematic reviews have shown a nearly sevenfold higher OA risk in ACL-injured knees, a 36% prevalence at 10 years after

ACLR, and no significant reduction in OA risk with operative versus nonoperative treatment²³. Ferrero et al. and Hohmann et al. reported higher OA rates after ACLR than after nonoperative care (risk ratio [RR], 1.72; 95% CI, 1.18 to 2.53)^{24,25}. Similarly, Komnos et al., Migliorini et al., and Everhart et al. also reported high OA rates in reconstructed knees²⁶⁻²⁸, with Everhart et al. reporting increased risk versus that in contralateral knees.

Studies evaluating addition of LET augmentation to ACLR have shown improved patient-reported outcome measures (PROMs), namely Lysholm and Tegner scores, supporting treatment of rotatory instability to reduce residual laxity²⁹. However, concerns remain that LET may

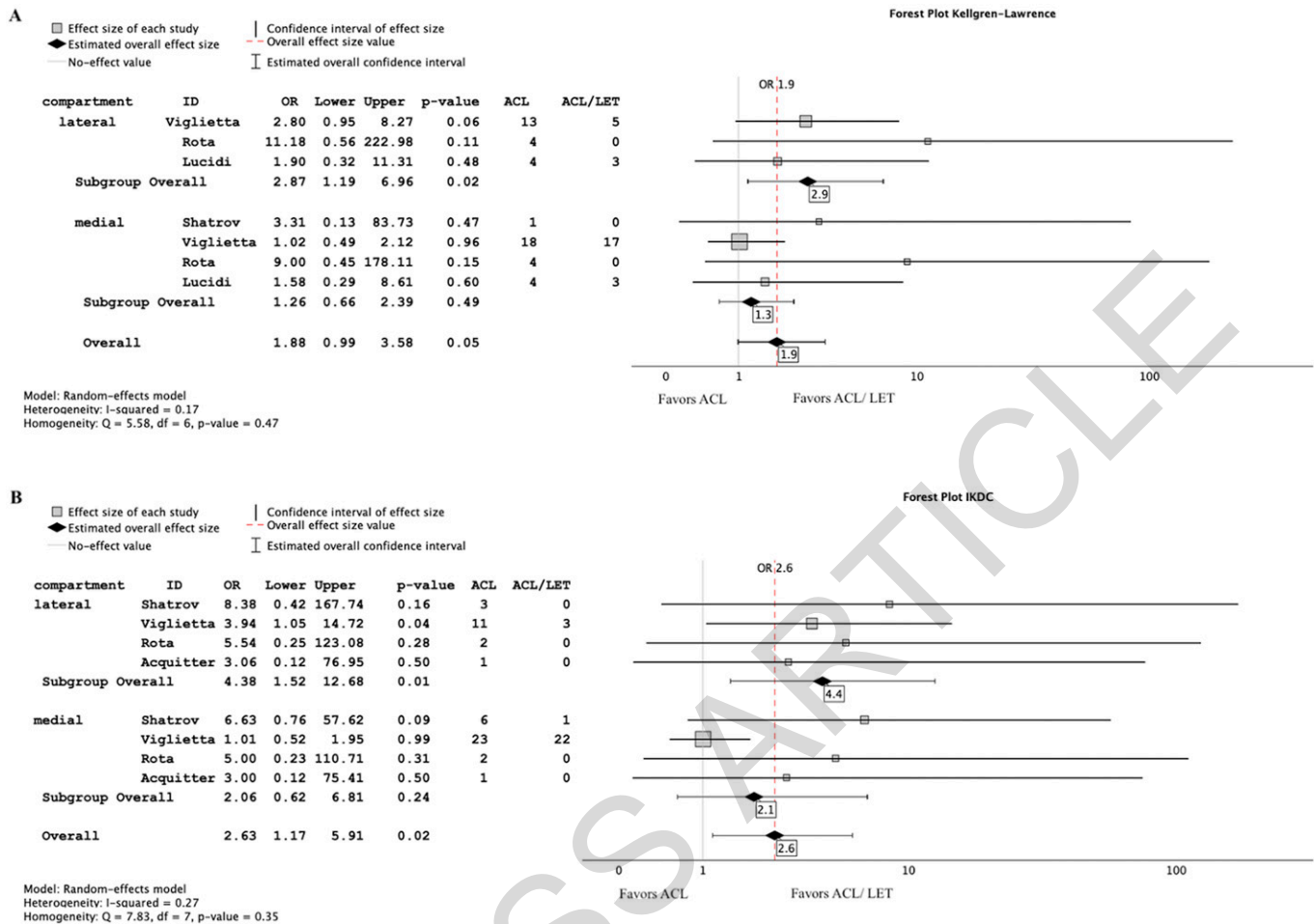


Fig. 3

Forest plots showing the odds ratios (ORs) for the development of moderate-to-severe osteoarthritis (OA) in the lateral and medial compartments after ACLR with versus without LET according to the Kellgren-Lawrence classification system (**Fig. 3-A**) and the International Knee Documentation Committee (IKDC) classification system (**Fig. 3-B**). df = degrees of freedom.

overconstrain the knee and accelerate lateral compartment degeneration^{30,31}. In contrast, our findings, which were consistent across classification systems, suggest that LET may reduce, rather than increase, lateral OA risk.

Supporting these findings, Marom et al.³⁴ demonstrated in vitro that even though LET increases lateral contact force under simulated pivoting loads, it also expands the contact area, thereby reducing contact stress. These biomechanical data align with our observation of a protective role for LET and reinforce its utility in managing anterolateral rotatory instability. Similarly, Delaloye et al.³² showed that both anterolateral ligament reconstruction (ALLR) and a modified Lemaire LET, when combined with ACLR, restored anterior translation and internal rotation to near-intact levels. Geeslin et al.³³ also found that LET significantly reduced tibial internal rotation and rotatory laxity when used with ACLR. These studies suggest that extra-articular procedures not only restore kinematics similar to those of the native knee but may also reduce the risk of secondary injury by enhancing joint stability.

Biomechanical studies of the effect of LET on lateral compartment contact pressure have reported mixed results^{34,35}. Shimakawa et al.³⁶ found no significant pressure change with LET, whereas others reported increased pressures with the modified McIntosh and Lemaire techniques^{37,38}. Notably, no pressure increases were seen with ALLR or the modified Ellison technique. The observed protection against OA may relate to improved kinematics³⁹, reduced residual laxity⁴⁰, and prevention of secondary meniscal tears and cartilage damage⁴¹. Still, additional high-quality clinical studies are needed.

Meniscal tears and meniscectomy are well-established OA risk factors⁴². Meniscectomy contributes not only through loss of tissue and altered load distribution but also by worsening knee instability. Performing concomitant meniscectomy with ACLR is associated with a higher incidence of OA within 5 years, whereas concomitant meniscal repair appears to reduce this risk⁴³. This likely reflects cartilage exposure and incomplete restoration of normal joint motion⁴⁴. Adding LET during ACLR may restore native-like mechanics and mitigate OA risk even when meniscectomy is required. In our study,

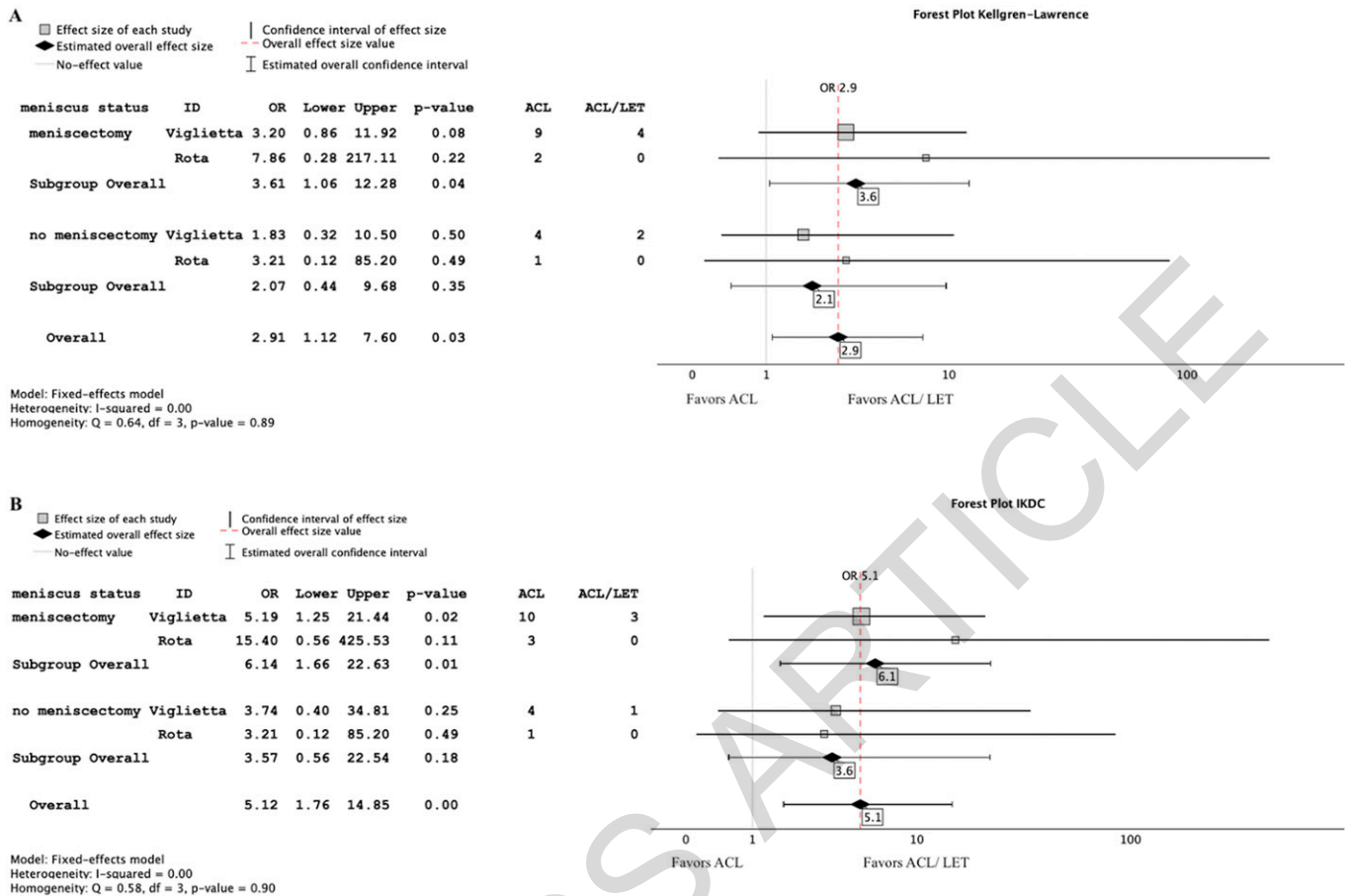


Fig. 4

Forest plots showing the odds ratios (ORs) for the development of moderate-to-severe osteoarthritis (OA) after ACLR with versus without LET in subgroups of meniscectomized and non-meniscectomized knees according to the Kellgren-Lawrence classification system (**Fig. 4-A**) and the IKDC classification system (**Fig. 4-B**). df = degrees of freedom.

addition of LET significantly reduced OA prevalence in meniscectomized knees. Özbek et al.⁴⁵ showed that partial medial meniscectomy increased laxity, while adding LET reduced anterior translation and internal rotation⁴⁵. Similarly, Gibbs et al.³⁹ found that ACLR+LET better restored native kinematics than ACLR alone in a clinical series of 20 patients with combined ACL and meniscal injuries.

Our study has limitations. First, the number of included studies was small (3 RCTs and 3 cohort studies), and the overall methodological quality was variable, ranging from low to good. Second, we could not assess the impact of graft type, tunnel placement, LET technique, or amount of meniscal tissue removed, due to limited data. Third, preoperative OA status was inconsistently reported. Fourth, restricting the analysis to articles in English, German, and French may have introduced language bias. Fifth, all studies were from Europe, possibly introducing selection bias. Finally, activity level, genetics, and surgeon experience may also influence OA. However, our study has several strengths. It included a comprehensive literature search focused on radiographic progression to moderate-to-severe OA at adequate follow-up durations (5 to 20 years). To our knowledge, this is the first meta-analysis

addressing the long-term impact of adding LET to ACLR on OA.

Conclusions

This systematic review and meta-analysis suggests that adding LET to ACLR may reduce the long-term risk of moderate-to-severe OA, particularly in the lateral compartment and in patients who have undergone meniscectomy. These findings, which were observed across multiple OA classification systems, support the potential role of LET in surgical decision-making, especially in patients at elevated risk for posttraumatic joint degeneration due to meniscal deficiency.

Appendix

eA Supporting material provided by the authors is posted with the online version of this article as a data supplement at jbjs.org (<http://links.lww.com/JBJS/1988>). ■

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